

CRACK-MORPHOLOGICAL ASPECTS IN FRACTURE MECHANICS

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Abstract—Fracture mechanics can be regarded as a methodology to characterize the actual crack by using the parameters given by the analysis of a simple mathematical crack model. Therefore, the selection of these crack models has very significant meanings. In recent years, several precise crack models have been proposed.

In this paper, we paid attention to the influences of the two-dimensional crack geometry and presented the numerical value of the elastic analytical solutions for the non-linear shaped cracks, those are, the bent, the curved and the branched crack. On the basis of these results, the characteristics of these cracks were discussed and the methods of approximate replacement to a simple straight crack were proposed.

We examined a zig-zag growth of crack and the crack branching observed in actual fracture processes.

1. DEFINITION AND SIGNIFICANCE OF CRACK-MORPHOLOGY

IN A SENSE, fracture mechanics can be regarded as a methodology to characterize the actual crack behaviors by using the parameters given by the analysis of some simple mathematical crack models. The crack models generally used are limited to the simplest crack, that is, a straight line crack.

The purpose of this paper is to consider the behaviors and the characteristics of the non-linear shaped cracks, paying attention to the complicated crack configurations, those are, a bent, a branched, a curved and a zig-zag crack, generally observed in the macroscopic or microscopic crack growth or propagation processes. To fracture mechanics, we introduced one of the approaches which make the crack model precise, in the meanings that such non-linear crack configuration should be taken into consideration. We define this approach as "crack morphology" or "crack morphological approach".

Recently, in order to explain some particular phenomena, some approaches for making the crack model precise have been proposed. These approaches can be divided into the following problems from our viewpoint.

- (i) non-linear deformation
- (ii) partial closure of the crack surfaces
- (iii) non-linear shaped crack
- (iv) non-linear crack front
- (v) plural interacted cracks
- (vi) heterogeneity, composite.

In this paper, the problem (iii) is discussed and the problem (v) was reported in another paper [1].

Crack morphology seems to have significant meaning in the following cases. For example, the macroscopic branched cracks are often observed in stress corrosion cracking and crack propagation in brittle fracture. The brittle crack propagation and fatigue crack growth in the mixed mode stress intensity factor (SIF) state have often been discussed. Generally in these cases a crack does not follow a straight path, but a curved or a bent path.

There are many trials for combining micro and macro fracture mechanics. Also, in these cases the crack morphology seems to have significant meanings, because a microscopic crack usually has a complicated geometry, for example, the concave and convex crack front, the zig-zag crack path or branching.

2. SOME BASIC CHARACTERISTICS OF NON-LINEAR SHAPED CRACKS IN LINEAR FRACTURE MECHANICS

2.1 Introduction

Only a few reliable solutions for the stress intensity factors (SIF) of the non-linear shaped cracks have been obtained for some particular cases, and this fact has probably been a large trouble in the researches of crack morphology.

Some solutions for the stress intensity factors of the non-linear shaped cracks in the state of longitudinal shear were given by Sih[2], Nakayama[3], Smith *et al.*[4]. However, the analytical solutions for the SIF of the non-linear shaped crack in the two-dimensional stress states, which are more important to crack morphology, have scarcely been found out. Generally in this case, both the stress intensity factors of mode *I* and mode *II* appear. And for this reason, the analysis is not easy and many interesting problems can be expected to be found.

In this paper we restrict our discussion to the problem of the non-linear shaped cracks in the two-dimensional elastic stress state, in particular, under uniaxial uniform tension.

Recently, a couple of the experimental or numerical analyses for these kinds of problems have been reported[5]. Andersson[6] solved this problem on the basis of Muskhelishvili's method, but there were the fatal mistakes in his paper and the use of his numerical results is questionable[7, 8].

Taking his mistakes into consideration, we constructed the general analytical method of solving the SIF of the non-linear shaped crack in an infinite isotropic homogeneous plate in the two-dimensional elastic stress state[10], including a polynomial approximation and the truncation procedure[9] of a conformal mapping function. And we obtained the numerical results for various cases. The details of this method are reported in another paper[10] and only its outline is introduced in "Appendix" of this paper. In this paper, we showed the numerical values of the SIF of the bent, the branched and the curved cracks, and referring to our experimental observations and some results reported by other investigators, we summarized the basic characteristics and behaviors of these cracks. Besides, on the basis of these results, we tried to make a crack morphological approach to some particular phenomena, those are, the zig-zaging in fatigue crack growth and the branching of the crack in stress corrosion cracking.

2.2 *Some characteristics of a bent crack*

In order to clarify the sharp bent and the zig-zag behaviors of the crack, we proposed a basic crack model shown in Fig. 1, which is called "bent crack" in this paper.

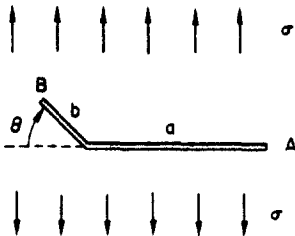


Fig. 1. Geometry of a bent crack in an infinite plane.

(a) *Stress intensity factors at the crack tip A.* Some of the SIF at the crack tip A of the bent crack in Fig. 1 (K_{1A} , K_{2A} or F_{1A} , F_{2A}) for $\theta = 45^\circ$ are shown in Table 1, normalized by the SIF for

Table 1. Stress intensity factors at the crack tip A of the bent crack for $\theta = 45^\circ$

b/a	F_{1A}	F_{2A}
0.01	1.000	0.003
0.1	0.998	0.019
0.2	0.995	0.028
0.5	0.988	0.033
0.8	0.985	0.024
1.0	0.985	0.016
1.5	0.988	-0.004
2.0	0.993	-0.022

$$F_{1A} = \frac{K_{1A}}{\sigma \sqrt{\pi(c/2)}}$$
$$F_{2A} = \frac{K_{2A}}{\sigma \sqrt{\pi(c/2)}}$$

a straight crack with the crack length c projected in the direction perpendicular to the tensile axis.

$$c = a + b \cos \theta \tag{2.1}$$

Table 1 shows that we can approximate K_{1A} and K_{2A} as follows,

$$K_{1A} \doteq \sigma \sqrt{\pi(c/2)}, \quad K_{2A} \doteq 0. \tag{2.2}$$

This relation is also applicable to the case of “forked crack”, which will be stated later, and we can say as follows. The stress intensity factor at a tip of the crack element, which is placed in the direction normal to the tensile stress, is not much influenced by the configuration of another crack tip, so far as these crack tips do not draw near to each other, and can be approximated by that of a straight line crack normal to the tensile direction, as eqn (2.2).

(b) *Stress intensity factors at the crack tip B.* The stress intensity factors at the crack tip B in Fig. 1, normalized by that of a straight line crack with the crack length c , F_{1B} , F_{2B} , are shown in Table 2, Figs. 2~4. Figure 4 shows that, with the increase of the bent angle θ , the F_{1B} value decreases and the absolute value of F_{2B} increases. Also Figs. 2 and 3 show that the values of F_{1B} and F_{2B} become almost constant, as the value of b/a increases. The ratios of K_{2B}/K_{1B}

Table 2. Stress intensity factors at the crack tip B of the bent crack for $\theta = 15^\circ, 30^\circ, 45^\circ, 60^\circ$

$$F_{1,2B} = \frac{K_{1,2B}}{\sigma \sqrt{\pi(c/2)}}$$

	$\theta = 15^\circ$		$\theta = 30^\circ$		$\theta = 45^\circ$		$\theta = 60^\circ$	
b/a	F_{1B}	F_{2B}	F_{1B}	F_{2B}	F_{1B}	F_{2B}	F_{1B}	F_{2B}
0.1	0.9540	-0.2120	0.8245	-0.3895	0.6339	-0.5053	0.4106	-0.5462
0.2	0.9496	-0.2346	0.8076	-0.4307	0.5983	-0.5578	0.3583	-0.5996
0.4	0.9466	-0.2555	0.7957	-0.4690	0.5741	-0.6072	0.3189	-0.6516
0.6	0.9457	-0.2648	0.7927	-0.4858	0.5679	-0.6283	0.3112	-0.6718
0.8	0.9456	-0.2694	0.7922	-0.4940	0.5678	-0.6376	0.3128	-0.6770
1.0	0.9457	-0.2718	0.7928	-0.4981	0.5694	-0.6413	0.3171	-0.6775
1.5	0.9463	-0.2737	0.7951	-0.5008	0.5744	-0.6414	0.3273	-0.6682
2.0	0.9468	-0.2733	0.7971	-0.4996	0.5785	-0.6377	0.3340	-0.6580

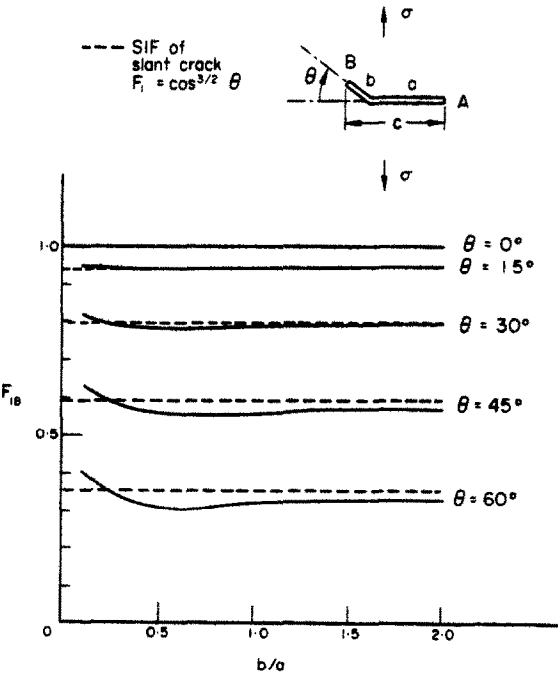


Fig. 2. Normalized stress intensity factors F_{1B} of the bent crack.

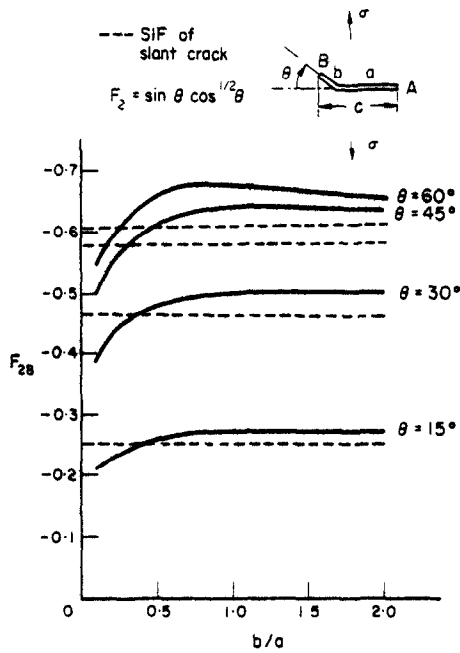


Fig. 3. Normalized stress intensity factors F_{2B} of the bent crack.

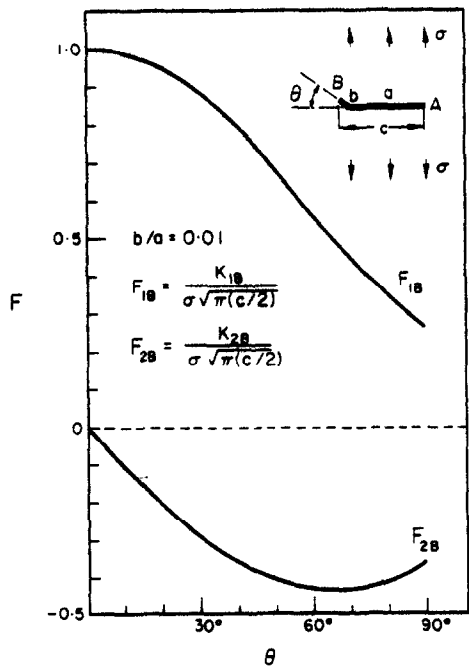


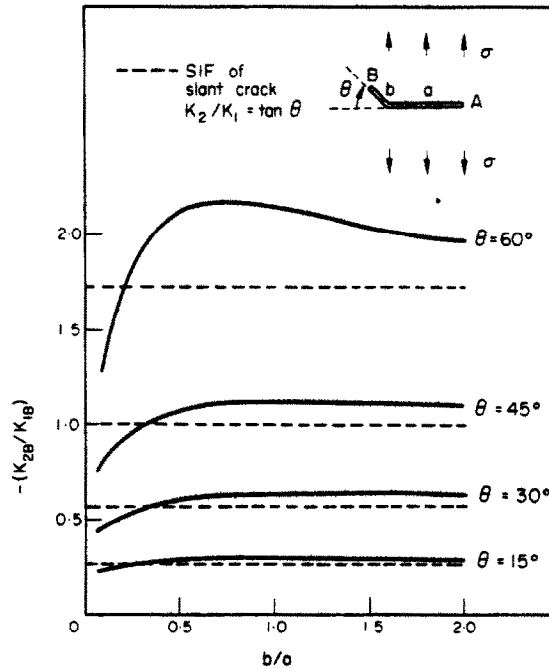
Fig. 4. Normalized stress intensity factors of the bent crack for $b/a = 0.01$.

calculated from Table 2 are shown in Fig. 5. The dotted lines in Figs. 2, 3 and 5 show the values of the SIF of the slanted straight cracks with the same inclination angle θ and with the same projected crack length c as the bent crack, K_1 , K_2 or F_1 , F_2 . They are given by the following equations.

$$F_1 = K_1/\sigma \sqrt{\pi(c/2)} = \cos^{3/2} \theta \tag{2.3}$$

$$F_2 = K_2/\sigma \sqrt{\pi(c/2)} = \sin \theta \cdot \cos^{1/2} \theta \tag{2.4}$$

$$K_2/K_1 = \tan \theta. \tag{2.5}$$

Fig. 5. Ratio K_{2B}/K_{1B} of the bent crack.

So far as $\theta = 0 \sim 45^\circ$, $b/a > 0.3$, the curves for the SIF at the crack tip B agree fairly well with the dotted lines and K_{1B} and K_{2B} can be approximately given by the following equations,

$$K_{1B} \approx \sigma \sqrt{\pi(c/2)} \cos^{3/2} \theta, \quad K_{2B} \approx \sigma \sqrt{\pi(c/2)} \sin \theta \cdot \cos^{1/2} \theta. \quad (2.6)$$

(c) *Approximate replacement of the bent crack to equivalent simple cracks.* Referring to the calculations stated above, a bent crack can be replaced by straight line cracks as shown in Fig. 6, which give nearly the same SIF. This suggests that the direction of crack at each tip and the crack

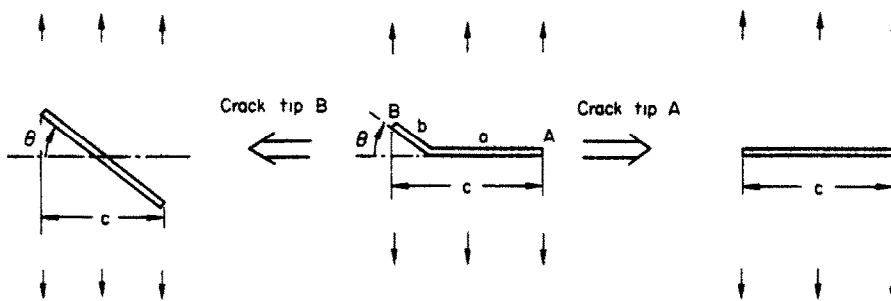


Fig. 6. Approximation methods for stress intensity factors of the bent crack.

length projected in the direction normal to the tensile axis give a predominant influence on the SIF. In other words, we can estimate the approximate value of the SIF of such crack, if the direction of crack at the tip and the projected crack length are given. But, this approximation is not applicable to the bent crack, the branch of which is relatively small enough as shown in Fig. 4, nor the curved crack stated later.

(d) *Behaviors of the bent crack.* As for the direction of crack propagation or growth from the crack tip with mixed mode SIF, several criteria or theories [11, 12] have been presented. But we cannot find significant difference among them. So, in the following discussion on the behavior of the bent crack, we took the theory of maximum circumferential stress (σ_θ) at the crack tip by Erdogan and Sih [11].

The direction of the crack growth, θ^* , is given by the following equation,

$$\frac{K_2}{K_1} (3 \cos \theta^* - 1) + \sin \theta^* = 0.$$

Substituting the SIF ratio obtained from Fig. 5 or Table 2 into this equation, the initial direction of the crack propagation or growth from the bent crack can be presumed as shown in Fig. 7. Figure 7

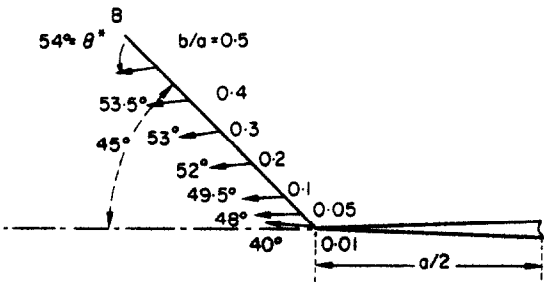


Fig. 7. The direction in which the bent crack propagates.

shows that the bent crack has a tendency to start to grow in the direction almost normal to the tensile axis, if b/a is small, and to start to grow pointing at the far extension of the initial crack, if b/a is large. We carried out the fatigue experiment of a mild steel sheet with a bent crack, shown in Fig. 8. The crack grew in the direction Fig. 7 suggested.

(e) *On the zig-zaging of crack.* The fatigue crack grows more or less in the manner of zig-zaging. This tendency seems to be more evident in some metallic materials such as aluminum alloys, copper alloys or stainless steels. Figure 9 shows such a kind of fatigue crack in a pure copper sheet. Besides, remarkable zig-zaging cracks were observed in stress corrosion cracking and corrosion fatigue processes.

Simply deducing from Fig. 7, it seems difficult to explain the zig-zaging crack by the model in Fig. 10(a) which was seen in *B. Cotterell's* paper[13]. And it seems more natural to consider the zig-zaging crack as a stochastic process shown in Fig. 10(b). That is, we consider that the crack path is easily bent or branched by the metallurgical effects and the others within 20 or 30 degrees of bent or branch angle and a small distance, as shown in Fig. 10(c), because in such region the K_1 values do not change so much and the K_2 values of bent crack are very small as shown in Fig. 4. We call it "Snaking Crack Growth". Then, why does not the crack path deviate in one direction? It is explained as follows. If the crack is bent by some causes, it will be bent again in the direction almost normal to the tensile stress by the effects of K_2 . And then it will be bent again from a certain

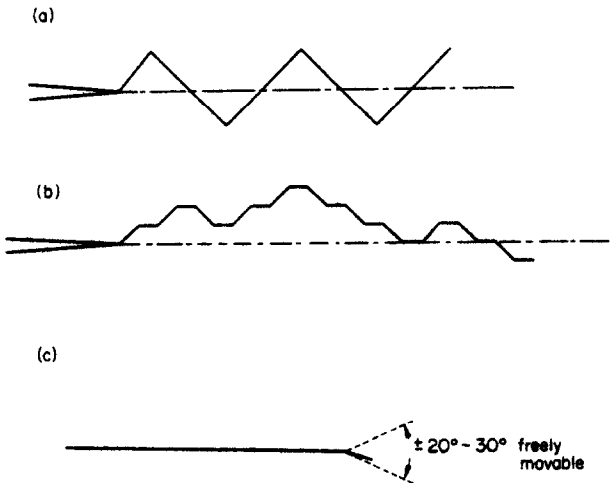


Fig. 10. The sketch of the zig-zaging crack.

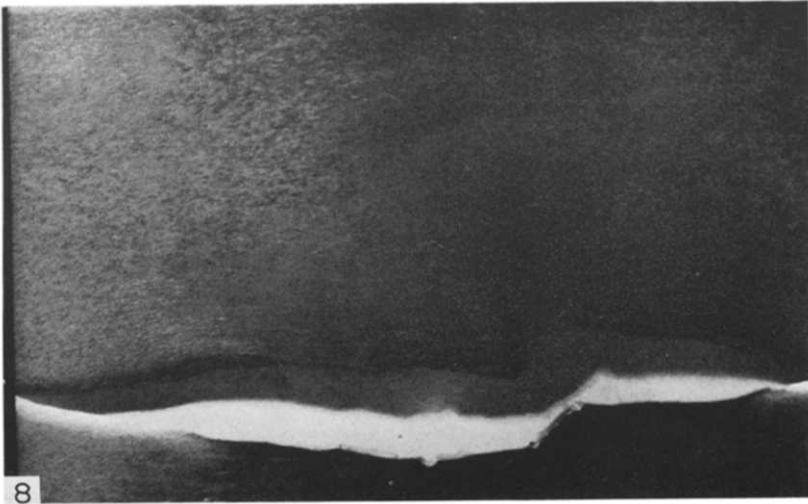


Fig. 8. Fatigue test of the bent crack in mild steel.

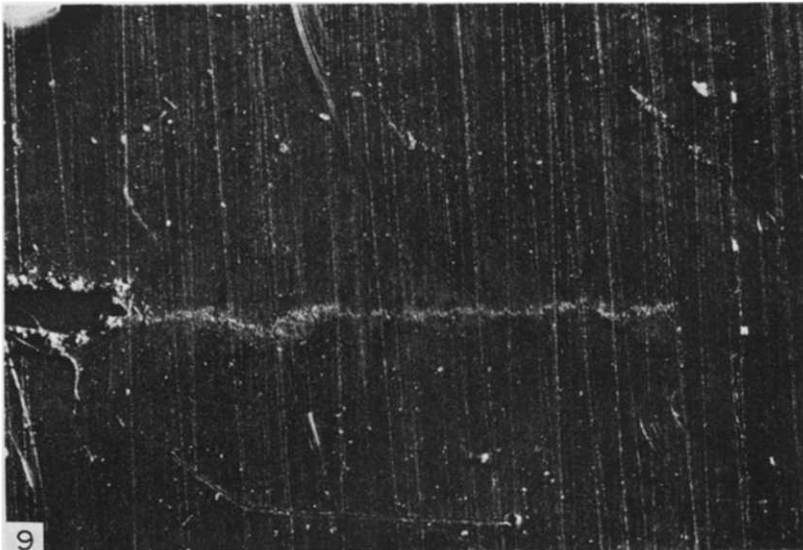


Fig. 9. Fatigue crack in copper.

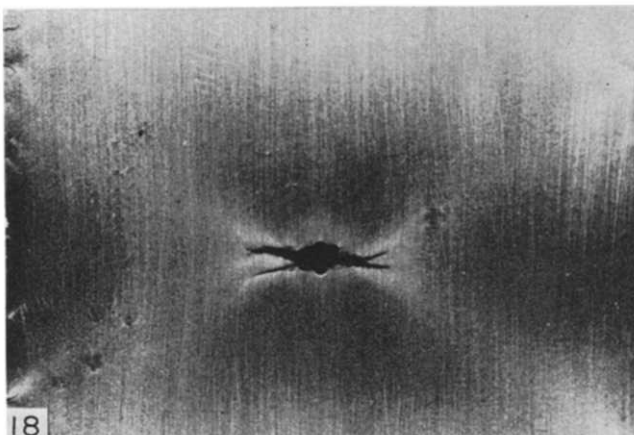


Fig. 18. Branched crack in stress corrosion cracking in stainless steel.

point in a certain direction, which is statistically determined. As shown in Fig. 10(b). If the crack route deviates remarkably apart from the prolongation of the original crack in any side, the crack will obtain a tendency to return back slowly. After all, it may be said that a macroscopic crack path can not grow much apart from the prolongation of the original crack. In practice, it is often observed that most of the directions of the short elements in a zig-zag crack are almost normal to the tensile axis and macroscopically the crack grows in that direction. In this case, the macroscopic crack growth rate seems to be not so much influenced by this zig-zaging of the crack path, and it seems to depend on the value of macroscopic K_I . The microscopic crack growth rate seems to be varied place to place by this zig-zaging.

2.3 Some characteristics of a curved crack

When we discuss the crack morphology, not only the bent crack composed of plural straight lines, which we mentioned above, but also the bent crack just as a curved line must be considered. We call it "Curved Crack". However, we can find out only a few analyses of such a curved crack. Sih and Paris[14] presented a formula of the SIF of a circular-arc crack by means of the stress function given by Muskhelishvili[14]. We computed the SIF for the two cases of such a crack referring to their formula. The results are shown in Figs. 11 and 12.

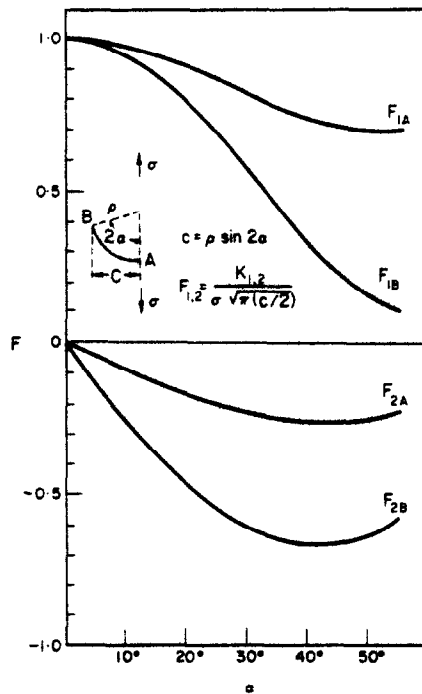


Fig. 11. Normalized stress intensity factors of circular-arc crack under uniform tension (No. 1).

The SIF at the crack tip A in Fig. 11, F_{1A} , where the tangent of the crack is perpendicular to the tensile direction, remarkably decreases from 1.0 and the absolute value of F_{2A} comes to a fairly large value, when α increases. This tendency seems to differ from that of the bent crack stated in 2.2(a). Therefore, in the case of a curved crack, in which the straight element at the crack tip is infinitely small, it is difficult to apply the same approximation method as that of a bent crack. The tendency of the results in Fig. 12 agrees fairly well with that of the results of the bent crack with small value of b/a shown in Fig. 4, when we put $\theta = \alpha$.

2.4 Some characteristics of a branched crack

The division of the crack tip into two or more branches is a well known phenomenon in the crack propagation in brittle fracture, stress corrosion cracking and fatigue crack growth. We call it "Forked Crack" here. Few reliable generalized analyses of the SIF of a branched crack have been obtained. We analysed a forked crack with two symmetric branches shown in Fig. 13.

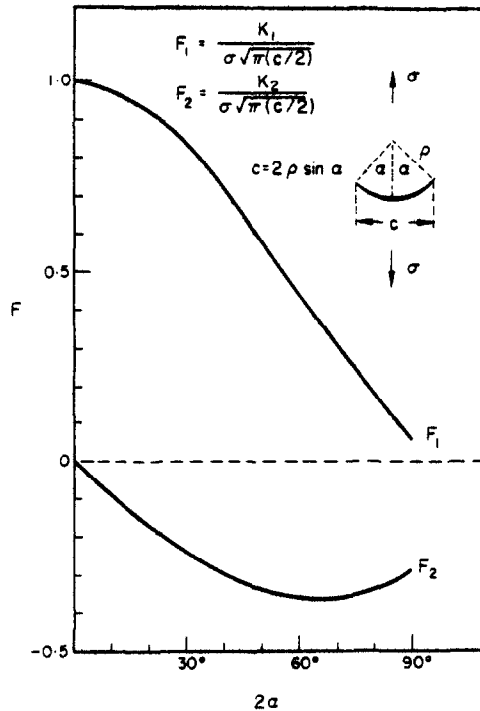


Fig. 12. Normalized stress intensity factors of circular-arc crack under uniform tension (No. 2).

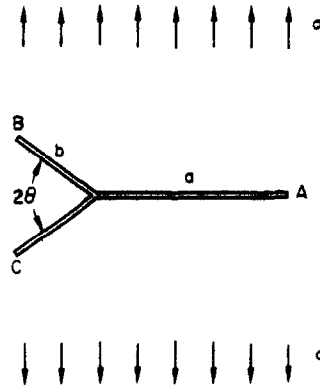


Fig. 13. Geometry of the forked crack under uniform uniaxial tension.

(a) *Stress intensity factors at the crack tip A of the forked crack.* The SIF at the crack tip A of the forked crack is normalized by that of a straight line crack with the crack length c and, as an example, the results for $\theta = 45^\circ$ are shown in Table 3. This shows that the characteristics of the bent crack which mentioned previously in 2.2(a) are applicable to this case too. And the following equations can be applied,

$$K_{1A} \approx \sigma \sqrt{\pi(c/2)}, \quad K_{2A} = 0. \quad (4.1)$$

(b) *The influence of a branching angle on stress intensity factors at the crack tip B of the forked crack.* The SIF at the crack tip B of the forked crack for $b/a = 0.1$ under uniform uniaxial tension were computed and the results are shown in Fig. 14. As the branching angle 2θ increases, then $F_{1B} (= K_{1B}/\sigma\sqrt{\pi(c/2)})$ values decrease. However, when 2θ was small enough, we could not obtain an accurate solution with a good convergence. But there seems to be a tendency that the K_{1B} value does not show a remarkable change and it is approaching to $1/\sqrt{2} \cdot \sigma\sqrt{\pi(c/2)}$ within the region of the small branching angle. The K_2 value shows an interesting feature. That is, F_{2B} (i.e. K_{2B}) equals 0 where $2\theta = 30^\circ \sim 40^\circ$. This is almost the same result as that given by

Table 3. Stress intensity factors at the crack tip *A* of the forked crack for $\theta = 45^\circ$

b/a	F_{1A}	F_{2A}
0.1	1.010	0
0.3	1.019	0
0.5	1.026	0
0.8	1.037	0
1.0	1.044	0
1.5	1.062	0
2.0	1.079	0

$$F_{1,2A} = \frac{K_{1,2A}}{\sigma \sqrt{\pi(c/2)}}$$

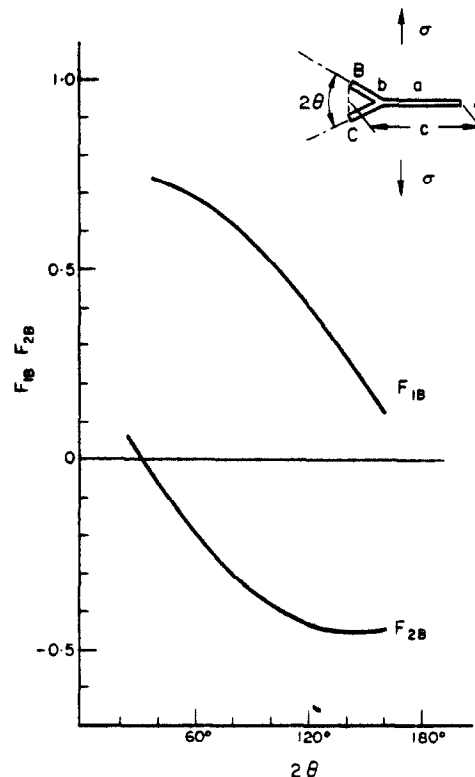


Fig. 14. Normalized stress intensity factors of the forked crack for $b/a = 0.1$.

Kalthoff[16] who analysed the branched edge crack in a finite plate by means of the finite difference method. Therefore, in regard to the symmetric branched crack under uniaxial uniform tension in an infinite plate, a similar tendency can be expected. If we assume that each branch can grow only in the direction of $K_2 = 0$, it can be well explained that the average branching angle observed in brittle fracture and stress corrosion cracking is almost $30^\circ \sim 40^\circ$.

(c) *The influence of a branch length on stress intensity factors at the crack tip B of the forked crack.* The SIF of the forked crack for $\theta = 45^\circ$ with respect to a branch length b are shown in Figs. 15 and 16. These SIF are normalized by that of the straight line crack with the crack length a , c , respectively. Figure 16 shows that, once the branching occurs, the K_{1B} value decreases promptly and then gradually increases, as the " b " increases. This tendency corresponds well to the result of a branched crack given by Kobayashi[17] by means of the finite elements method. Also, this tendency of K_1 exists even if θ is not 45° . Therefore, in the case of the fracture process in which the crack growth rate depends on K_1 , it can be said that once a branch occurs, the branching is accompanied with a temporary decreasing growth rate effect, a crack arresting effect or a constant velocity effect.

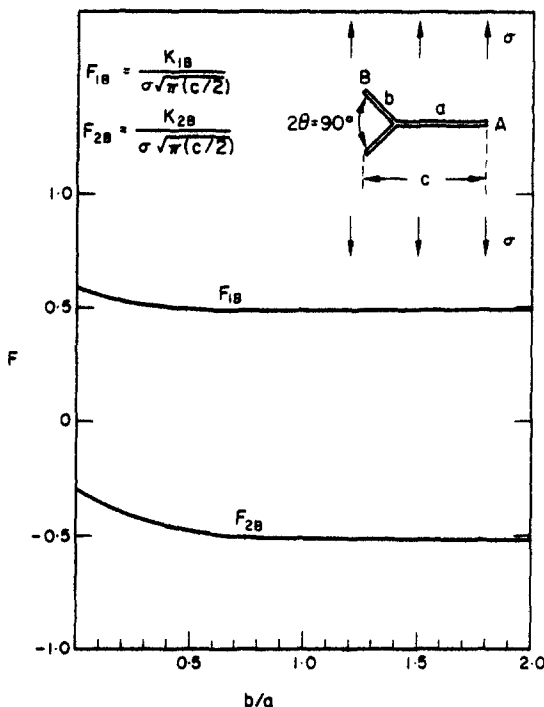


Fig. 15. Stress intensity factors of the forked crack normalized by SIF of crack length c .

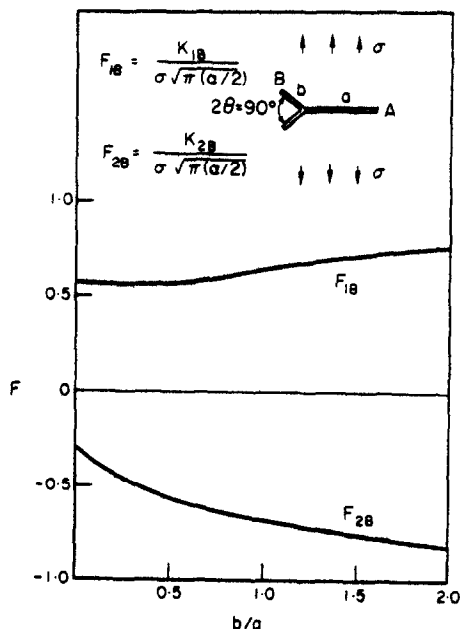


Fig. 16. Stress intensity factors of the forked crack normalized by SIF of crack length a .

(d) *Unstability of an asymmetric branched crack.* Theocaris analysed the asymmetric branched crack shown in Fig. 17 by means of an optical method[18]. These results show a tendency similar to that of the results of the bent crack or the forked crack and imply a very interesting fact as follows. That is, when one branch becomes longer than the other, the K_I value at the crack tip of the larger branch increases rapidly and that of the shorter one decreases fast. This means that there is a following possibility. In the case of a fracture process where the crack growth rate is sensitive to the change of K_I value, it is possible that, even if a microscopic branch occurs, a slight unbalance of K_I between the two branch tips accompanies. This can make one branch much longer and the K_I value at that crack tip larger, and on the other hand, the K_I value

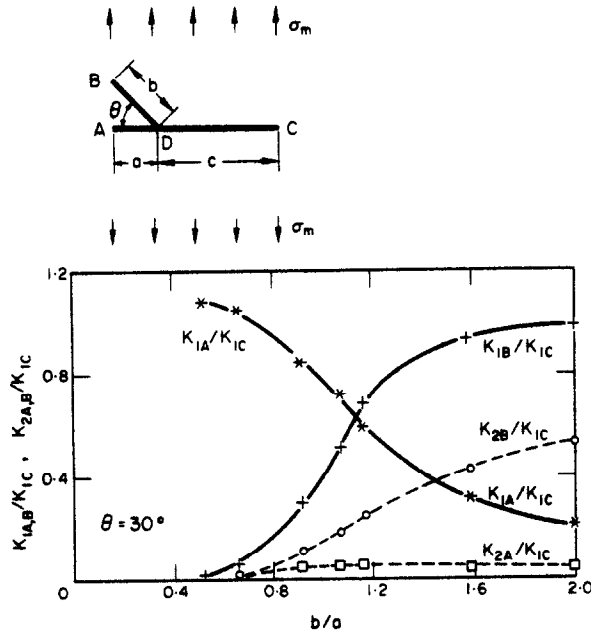


Fig. 17. Stress intensity factors and the geometry of the bifurcated crack (Theocaris[18]).

and the crack growth rate at the other branch tip much smaller. And the smaller branch will be left behind. That is, when the crack growth rate is sensitive to the change of K_I , it seems to be difficult that two or more macroscopic branch cracks can simultaneously grow. And such a macroscopic branched crack can be observed only in the case when the crack growth rate is not so sensitive to the change of K_I . It is probable that such a loss of the K -dependency in these phenomena is also brought by the successive microscopic branching and the crack growth rate microscopically depends on K_I . As a result, it may be said that, if this microscopic branching is followed in details correctly, the crack growth rate depends on K_I in most cases.

(e) *Branching in stress corrosion cracking.* The crack branching phenomenon in stress corrosion cracking has often been observed and the systematic researches have been carried out by Carter [19], Speidel [20] and others. Figure 18 shows a branched crack caused by controlling (preventing) the generation of the distributed cracks on the surfaces of the stainless steel specimens in the stress corrosion cracking tests [21]. These branching phenomena in stress corrosion cracking can be discussed in the followings from the viewpoint of crack morphology.

(i) *Determination of $K_{I\text{SCC}}$.* As Speidel has pointed out, when the $K_{I\text{SCC}}$ value is determined by the K -decreasing type tests, the higher value of $K_{I\text{SCC}}$ are often obtained, if we use the SIF of a straight line crack regardless of the branching. This is because the K_I value decreases due to the branching and the arresting of crack occurs at an apparent higher value of K_I than the correct value of $K_{I\text{SCC}}$, in such a test procedure that the test starts at higher level of K_I and the branches occur and then $K_{I\text{SCC}}$ is determined by decreasing K_I level. On the other hand, in the series of K -increasing type tests, $K_{I\text{SCC}}$ can be decided by the SIF of a straight line crack because the branch does not occur at the low level of K_I . In the K -decreasing type tests, it seems that we can obtain the correct $K_{I\text{SCC}}$ value if we can calibrate the value of SIF by the results mentioned previously.

(ii) *The conditions for the occurrence of the branching and the region that the crack growth rate is independent of K .* According to the researches by Speidel [20], Carter [19] and Congleton [22], it is said that the branching can occur when the SIF value comes to be equal or larger than 1.4 times or 2 ~ 4 times $K_{I\text{SCC}}$. And as for the cause of the branching, it is said that an influence of non-linear deformation at the vicinity of the crack tip, heterogeneity of crystal structures or distributed cracks can be considered.

Speidel distinguished the branching into two types, those are, micro- and macro-branching and assumed that each branching occurs in the region as illustrated in Fig. 19. Micro-branching is thought to be such a fracture process that, in the asymmetric branched crack growth mentioned before, shorter branch cracks are frequently left behind by a main crack. The branching angle of

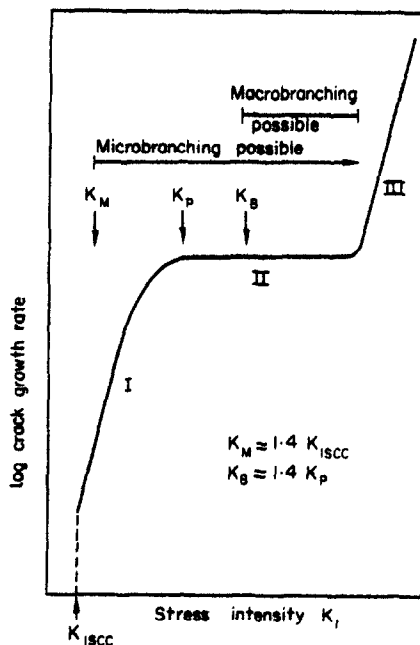


Fig. 19. Schematic representation of the dependence of SC crack growth rate, microbranching and macrobranching on the crack tip stress intensity (Speidel[20]).

micro-branching is larger than that of macro-branching and its value is randomly scattered. From a macro-scopic point of view, the micro-branching seems to follow a straight line path leaving behind many small branches. At the region I of the $K - da/dt$ diagram where the crack growth rate is sensitive to the change of the K_I value, it seems that the micro-branchings occur as mentioned above.

On the other hand, Speidel said that one of the necessary conditions for the occurrence of macro-branching is that the crack velocity versus stress intensity curve must be plateau. This corresponds to the argument by Clark and Irwin[23] that the velocity of a brittle crack at the stress intensities higher than the critical value is essentially constant before the crack division in brittle crack propagation.

If we accept the condition proposed by Speidel, we must search for the reason of the apparent loss of K -dependency in the process of micro-branching. It can be explained for an example as follows. If we assume that the crack tip divides when the K_I value becomes larger than a certain value, during the growing process of the branched crack, the crack tip of the one or both branches should divide again. If we can assume that such a second or third branching occurs, the K_I value and the crack growth rate da/dt will repeat the small amount of their increases and decreases, even if the apparent one crack length increases in one direction, as shown in Fig. 20. Therefore in the growing process accompanied by micro-branching, where the micro-branching and the growth are repeated, it can be said that the average crack growth rate remains almost constant as far as the interval of the generation of micro-branching satisfies a certain condition. It seems that this apparent constant growth rate caused by micro-branching is the reason for the macro-branching mentioned above.

In our stress corrosion cracking tests of a stainless steel, the crack growth rate varied violently and randomly as shown in Fig. 21 and, however, we obtained a plateau, i.e. the region II, by smoothing this irregular curve. In this case, crack growth is influenced by both the relatively small change of K_I and the large change of K_I including the former change.

Moreover, when we take it into account that the branching angle of macro-branching can be explained well by the K_2 of the forked crack mentioned before, we can say that there is K -dependency even in the region II of stress corrosion cracking where it has been said that the crack growth rate is independent on the change of K_I .

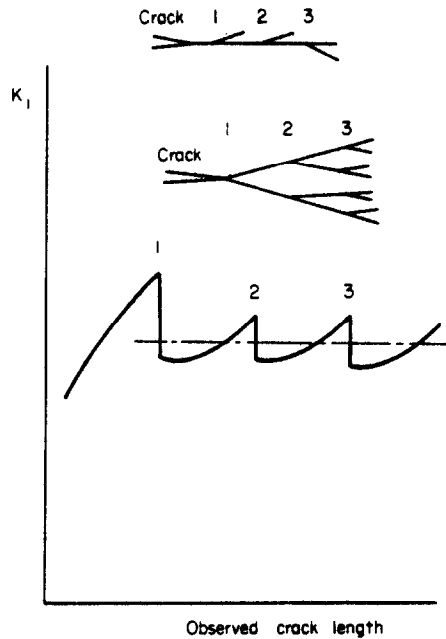


Fig. 20. Expected values of K_I for multifurcated crack vs observed crack length.

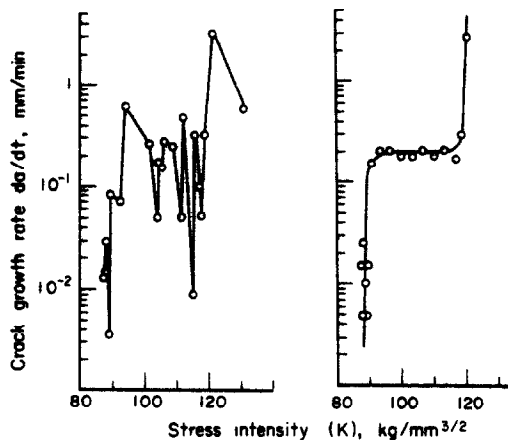


Fig. 21. Relation between crack growth rate and stress intensity factors in stress corrosion cracking of stainless steel.

3. SUMMARY

We analysed the SIF of the non-linear shaped cracks, those are, the bent, the curved and the branched crack, under the two-dimensional stress state and discussed the characteristics of these cracks on the basis of these results and taking the experimental results into consideration, we discussed the behaviors of these cracks.

We summarized some of the interesting points in the following.

(1) We examined the method or the possibility to replace these non-linear shaped cracks with a simple straight line crack equivalent in respect of the stress intensity factors.

(2) The zig-zaging crack path can be explained well by the characteristics of the bent crack and the stochastic process.

(3) It has been said that the behavior of the branched crack in stress corrosion cracking which is the matter of great concern is one of the K -independent phenomena. But we showed that it is possible to say that the K -dependency still exists even in this case if we choose some proper values of K , taking the branching into consideration.

We have obtained some fruitful results from a crack morphological approach and this approach can be applicable to the crack path problem in the mixed mode SIF state. But, for the reason of spacing this problem will appear in another paper.

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APPENDIX

Outline of the method to solve the stress intensity factors of the branched crack under the two-dimensional stress state

We have made the general method to solve the SIF of the crack shown in Fig. 22 under the two-dimensional stress state. This crack has n branches emanating from a common origin and with certain lengths and with certain angular intervals between each two neighbouring branches.

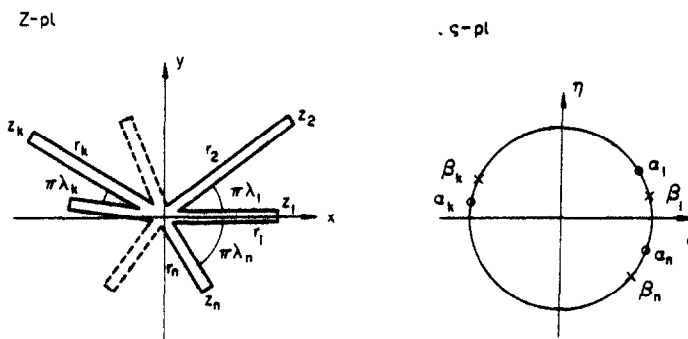


Fig. 22. Crack contour in original and in mapping plane.

Andersson[6] has solved the same problem by means of Muskhelishvili's[15] method, but there were fatal errors[7]. We corrected his errors and referred to the methods developed by Sih[14] and Bowie *et al.*[9], who utilized the conformal mapping function. This kind of method utilizing conformal mapping functions is often accompanied with the difficulties caused from the multi-valued function and asymmetry of the crack tip stress state. So we had not been able to find out a reliable generalized solution. It seems that this was one of the obstacles on the way of the development of crack morphological researches.

We constructed the analytical solution of these problems. And the details of this method will be published in another paper. Here we are going to explain merely the outline of this method.

The conformal mapping function is given by the following equation which transforms the configuration shown Fig. 22 into an unit circle,

$$\omega(\zeta) = A\zeta^{-1} \prod_{k=1}^n (\zeta - e^{i\alpha_k})^{\lambda_k} \quad (\text{A.1})$$

where, A is a real constant and α_k are the constants corresponding to the branching points. β_k shown in Fig. 22 are the constants corresponding to each crack tip. If the crack configuration is given, the mapping function can be determined by solving the non-linear equations[6].

The mapping function, eqn (A.1), is a multi-valued function and it is very difficult to determine the stress function directly by using eqn (A.1). So the eqn (A.1) is expanded in the series as shown in the following equation,

$$\omega(\zeta) = A \left(\zeta + \sum_{k=1}^n A_k \zeta^{1-\lambda_k} \right) \quad (\text{A.2})$$

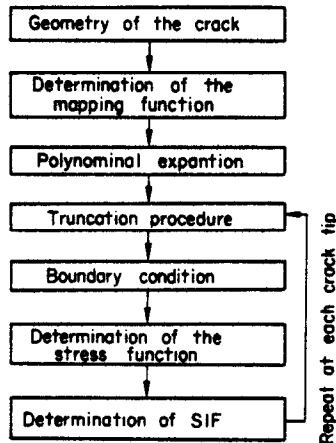


Fig. 23. Flow chart of the method.

where, A_k are some complex coefficients and determined by the recursive formula given by the following relation.

$$\omega(\zeta) \cdot \prod_{k=1}^n (\zeta - e^{i\theta_k}) = \omega'(\zeta) \cdot \prod_{k=1}^n (\zeta - e^{i\alpha_k}) \cdot \zeta. \quad (\text{A.3})$$

For practical reasons, the number of terms in the mapping function has to be kept finite. In this case, in order to preserve the crack tip geometry without disturbing the overall crack configuration, we adopted the truncation procedure by Bowie[9], making improvement on a couple of points.

When the stress function is determined, we used the method of polynomial approximation given by Muskhelishvili. We calculated complex stress intensity factors according to the following equation,

$$K = K_1 - iK_2 = \frac{2\pi^{1/2} \phi'(\gamma)}{[e^{i\delta} \cdot \omega''(\gamma)]^{1/2}}. \quad (\text{A.4})$$

Here, γ_i are the points on ζ -plane corresponding to the crack tips Z_i on the original Z -plane, and δ is the argument of Z_i . And $\phi'(\gamma)$ is the derivative of the stress function at the crack tip. We repeated this calculation process at each crack tip. Flow chart of this method is shown briefly in Fig. 23. On the basis of this flow chart, we have made the computer program which makes it possible generally to analyse such a kind of crack of various configurations. The program included several mathematical procedures, those are, a calculation of complex functions, a numerical solution of linear equations and others.